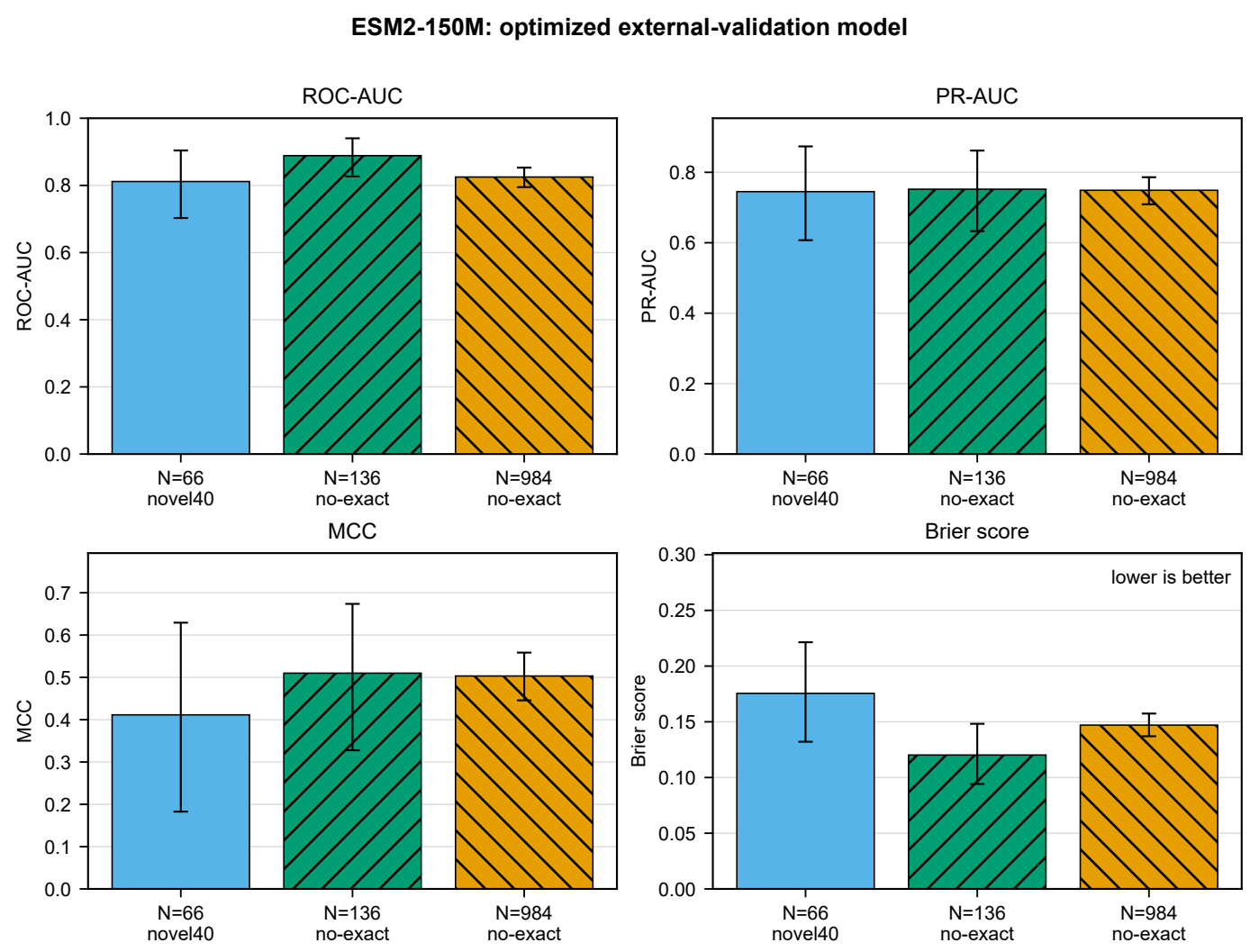


Note: The ESM2-150M column in Fig 4A is from a fixed-hyperparameter controlled ablation and differs from these optimized values by ≤0.045 across all metrics and panels.